

Learned Scale Modulation

Method 4 — Learned Scale Modulation

1. Plan

Learn a spatially varying correction to frozen DAV2 using RGB information, instead of a fixed affine mapping:

DFC RGB + DAV2
↓
small learned correction head
↓
scale/residual correction
↓
predicted AGL

Evaluation uses the same 4-fold spatial holdout protocol as Method 3.

2. Data / model / paths

DFC RGB:

data/dfc2019/raw/RGB/Track1-RGB/

DFC AGL truth:

data/dfc2019/raw/Truth/Track1-Truth/*_AGL.tif

DAV2 predictions:

data/dfc2019/experiments/dav2_baseline/depth/

Method 4 outputs:

data/dfc2019/experiments/method4/

Per-fold model checkpoints:

- **method4_fold0.pt**
- **method4_fold1.pt**
- **method4_fold2.pt**
- **method4_fold3.pt**

Final results:

data/dfc2019/experiments/method4/
method4_spatial_cv_results.json

3. Test — 4-fold spatial holdout

Each fold trains on 3 spatial quadrants and tests on the held-out quadrant. All four folds use the same 50 benchmark tiles.

Overall result:

Metric	Baseline	Method 4
MAE	3.8817 m	3.2663 m
RMSE	5.3409 m	5.1736 m
Pearson	0.5876	0.5469
Spearman	0.5112	0.4852

Fold-wise MAE improved in all 4 folds; Pearson and Spearman decreased in all 4 folds.

Fold results:

- Fold 0: MAE 3.6483 → 2.9028
- Fold 1: MAE 3.6298 → 3.2616
- Fold 2: MAE 4.5594 → 3.8493
- Fold 3: MAE 3.6892 → 3.0514

4. Conclusion

Method 4 is promising but not yet accepted as the final method.

It consistently reduces absolute elevation error, but consistently damages Pearson/Spearman correlation.

Therefore:

Method 4 → retain as current best experimental direction
--

Next step → modify the objective/model to preserve depth structure/rank
